# Tīwesdæg v1.1.2

created by k.odk

# Made to be played

Tīwesdæg reads two sound files as oscillators. A short window slides over each file and is read again and again — that repetition is the pitch. Move the window and the pitch moves.

The two readers are wired into each other. They cross-modulate, two comparators turn them into square waves, and two eight-bit shift registers clock each other from those squares — so the pattern never settles, and yet it never wanders off at random. Nothing in the sound uses random numbers: the same settings give the same behaviour.

It is made to be played, a little like a modular patch that lives in your hand. Around the circuit sit two pianos and a pad that take their tempo from the sound itself, five effects with a step sequence, and a photograph that can be read as voltage.

**Too much to learn? Skip it.** Tap the preset numbers on the right, one after another, and listen. Then load any sound into a deck — a voice, a room, a record — and it becomes the oscillator. Turn **TRIG** and the two **PEAK** rings slowly; that is most of the instrument.

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# What it needs

Tīwesdæg reads two files, runs a feedback circuit, two pianos, a pad and five effects in real time. It asks more of the device than a typical app.

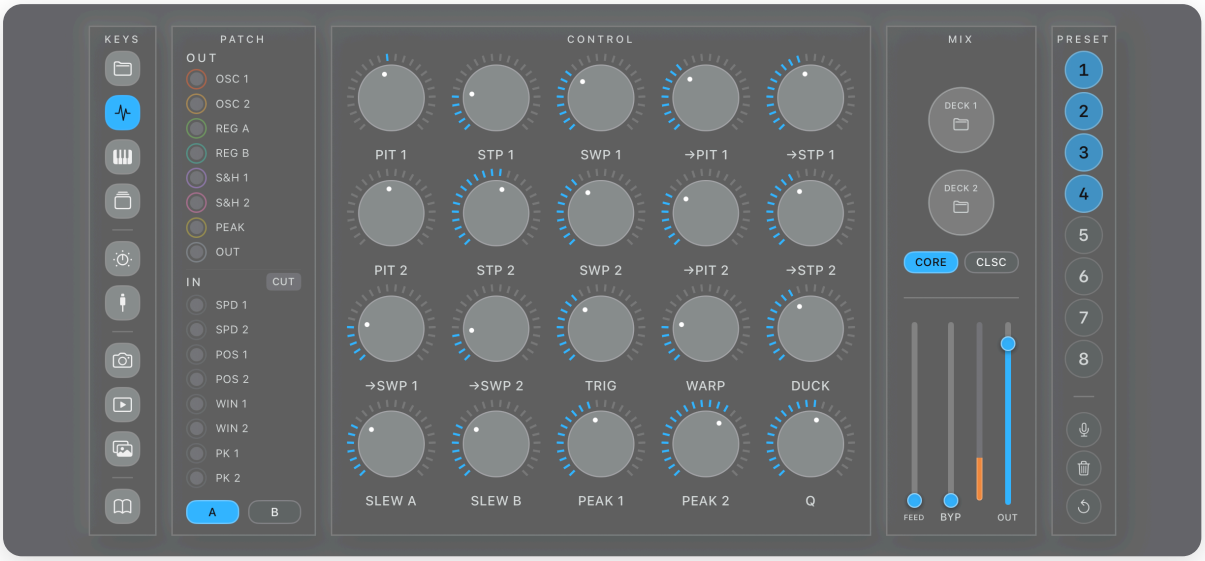
## System

| | |
|---|---|
| IOS | iOS 18.0 or later. |
| DEVICE | iPhone, played in landscape. **iPad:** the iPad layout is ready and will arrive in the next update (see section 12). |
| RECOMMENDED | iPhone 13 or newer. Tīwesdæg is developed and tested on an iPhone 13. |
| CAMERA | The back camera, for a single still frame (IMAGE). The frame is never saved to disk or to your photo library. |
| MICROPHONE | Only when you record into a deck. |
| FILES | Any audio file you can open from Files can be loaded into a deck. |

## About AUv3

Tīwesdæg is a **standalone app**, not an AUv3 plug-in. The camera and the microphone are part of the instrument, and a plug-in inside a host cannot use them in the same way.

# Keys, holes, rings FRONT




### KEYS — the left column

Folder: **SETTING** (decks and mixer). Waveform: **PEAK**, the two peaks (lit while they are in the mix). Piano keys: **PIANO**. Stack: **PAD**. Dial: **FX**. Plug: **AUDIO**. Camera: take one frame (hold to drop it). Play: run the red line over the picture. Photo: **IMAGE**. Book: the guide inside the app.

### PATCH

OUT on top, IN below — section 04. Under them, **A** and **B** close each shift register into a loop, so its pattern keeps coming round; open it and new bits keep coming in.

### CONTROL — twenty rings

The first ten belong to the two decks. Their names follow what the deck is: as an oscillator they read **PIT**, **STP**, **SWP**; reading a file they read **SPD**, **PLC**, **WIN**. A ring with an arrow (**→PIT 1**, **→STP 2** …) is how much the *other* side moves it.

**TRIG** is the comparator's hysteresis, and also the ceiling on how fast the registers turn. **CROSS** — or **WARP** when both decks are oscillators — is how much each side ducks the other, **DUCK** how quickly. **SLEW A/B** smooth the registers' steps. **PEAK 1**, **PEAK 2** and **Q** set the two resonant peaks.

### MIX and PRESET

**DECK 1**, **DECK 2** open a file. **CORE** is the current oscillator, **CLSC** the earlier way it sounded. **FEED** sends the output back into the readers; **BYP** lets sound pass around the two peaks; the orange bar is the level; **OUT** the volume. On the right: presets 1–8, the microphone, the bin and undo (section 11).

# Who sends, who receives

The holes carry the circuit back into itself. Eight outputs, eight inputs, one cord per input.

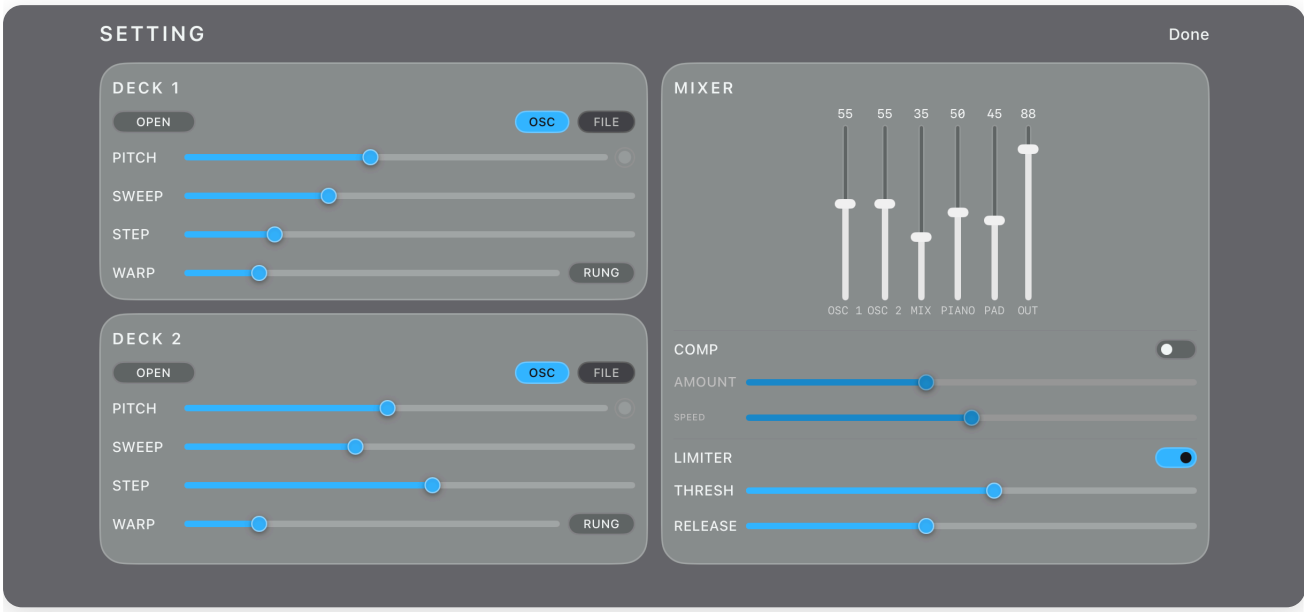
| OUT — what sends | | IN — what listens | |
|---|---|---|---|
| **OSC 1 · 2** | the two readers | **SPD 1 · 2** | how fast each head runs |
| **REG A · B** | the two shift registers, as stepped voltages | **POS 1 · 2** | where each window sits |
| **S&H 1 · 2** | sample-and-hold of the circuit | **WIN 1 · 2** | how long each window is |
| **PEAK** | the two peaks | **PK 1 · 2** | the two peaks |
| **OUT** | the whole sound | | |

## Making a cord

Tap an OUT hole to arm it, then tap an IN hole: a cord is strung. Each input takes one cord. **CUT** and then a tap pulls a cord out. An oscillator into its own window length, a register into a peak, the output into the speed — loops are the point.

The other panels have the same holes in the same colours. On PIANO, PAD and FX a short **CV** column lists REG A/B, S&H 1/2 and OSC 1/2: arm one there, then tap the small circle at the end of any slider to drive it. IMAGE adds four outputs of its own (section 10).

# The decks and the mixer SETTING




## DECK 1 · DECK 2

**OPEN** loads a file into the deck; at first both hold a triangle wave made by the app. **OSC** / **FILE** chooses what the deck is. As an oscillator: **PITCH**, **SWEEP**, **STEP** and **WARP**, with **RUNG** beside it to tie the steps to the registers. As a file reader: speed, window and place — the quarter/half/1/2/4 tags snap the speed, HOLD freezes the head, SWEEP stretches the window into a slow scan.

## MIXER

Six faders: **OSC 1**, **OSC 2**, **MIX** (the peaks), **PIANO**, **PAD** and **OUT**. Under them **COMP** (AMOUNT, SPEED) evens out loud and quiet, and the **LIMITER** (THRESH, RELEASE) holds the ceiling with a short look-ahead — the last thing the sound passes. Neither is there to colour the sound.

# A score that comes from the sound

PIANO




## The score

**NEW no.2** builds a score of thirty-two steps: one in ten a rest, one in ten a leap, the rest a step or two away, and every eighth step returns near the middle so a shape can be heard. It is made by turning a seed, never by rolling dice — the same number gives the same score. Both pianos and the pad read from it.

## MODE

**LINK**: the right side follows the left, with only DECAY and DETUNE free. **ALT** hands the score to one side at a time, so one line splits between the two and drifts apart. **COPY** and **PHOTO** take the line from elsewhere — PHOTO from the picture (section 10).

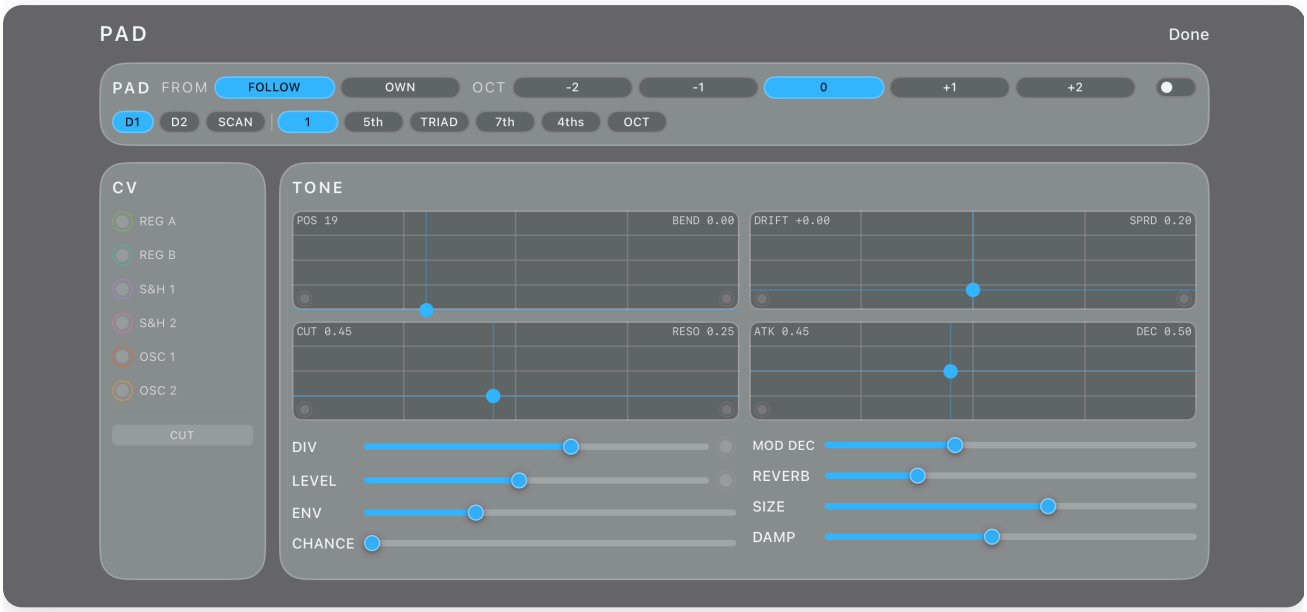
## TUNE and SCALE

TUNE cuts the octave into 12, 19, 24 or 31 equal steps; SCALE picks which steps are used. The last six — **ALPHA**, **BETA**, **GAMMA**, **BP**, **PHI**, **STRETCH** — never come back at the octave and sound their own cents. One choice covers both pianos and the pad.

## LEFT · RIGHT

Each side strikes when the comparator has fired **DIV** times, so the tempo comes out of the sound itself. Five tones — PIANO, BELL, MARIMBA, GLASS, GAMELAN — then **ROOT**, **RANGE**, **DECAY**, **DETUNE**, **LEVEL**. The switch at the top right turns each side on.

# Cycles lifted from your sound PAD



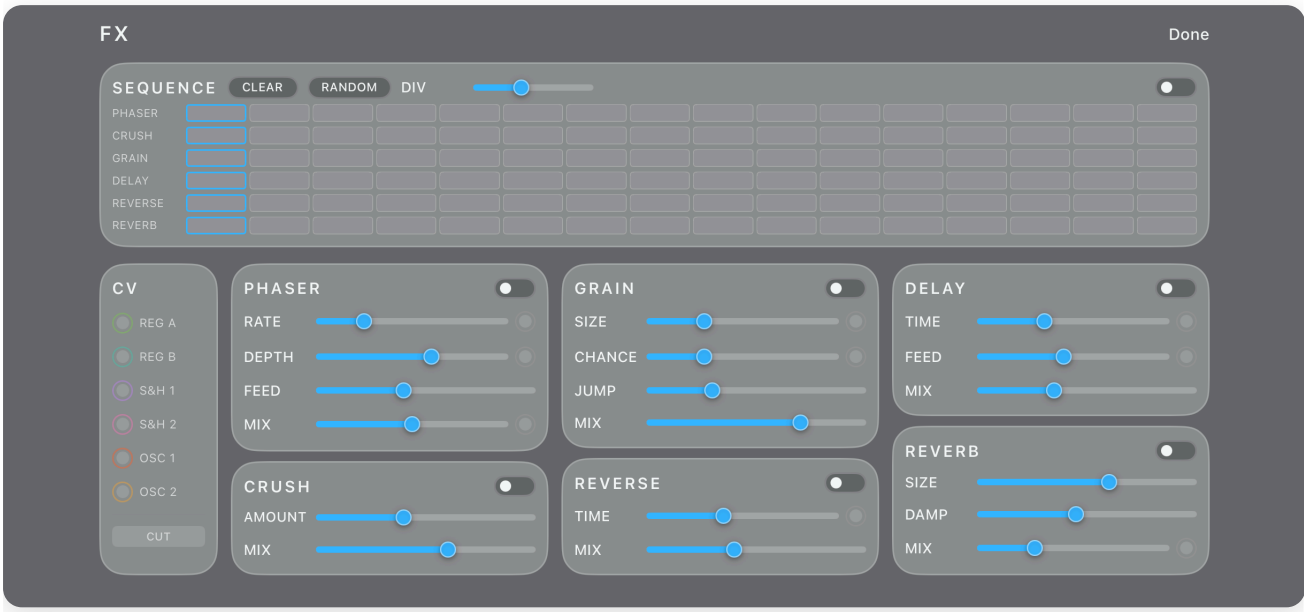

**SCAN** takes the chosen deck (**D1** or **D2**) — the file, or the oscillator when no file is in it — finds the pitch at sixty-four places through it and lifts one cycle from each, so the table walks through the timbres of that sound.

**FROM**: FOLLOW reads the piano's score, OWN its own. **OCT** shifts it. The chord row stacks the note in degrees of your scale — **1**, **5th**, **TRIAD**, **7th**, **4ths**, **OCT** — so a chord stays in that tuning. It changes every **DIV** clocks; **CHANCE** now and then steps off the score.

The four squares hold two settings each, one finger for both: **POS** (where in the table it reads) and **BEND** (bows the way one cycle is read, pushing harmonics forward); **DRIFT** (moves through the table at each note and settles back) and **SPRD** (a second, detuned read); **CUT** and **RESO**; **ATK** and **DEC**.

Below: **LEVEL**, **ENV**, **MOD DEC**, and a reverb (**REVERB**, **SIZE**, **DAMP**).

# After everything else FX




## The boxes

In this order: **PHASER**, six stages swept by its own slow wave; **CRUSH**, fewer steps read less often; **GRAIN**, a short piece grabbed and repeated one to six times, sometimes at another pitch; **DELAY**; **REVERSE**, a delay that fills and reads backwards; **REVERB**. Each has its own switch, and a box that is off costs nothing.

## SEQUENCE

Sixteen steps by six boxes. A step advances when the comparator has fired **DIV** times, and only the boxes marked on that step are in. **RANDOM** goes further: every step, all the knobs of the boxes are replaced — a different phaser, a different room, one step to the next. A seed is turned, so the same round comes back the same. **CLEAR** empties it.

# Where it comes in and goes out

AUDIO



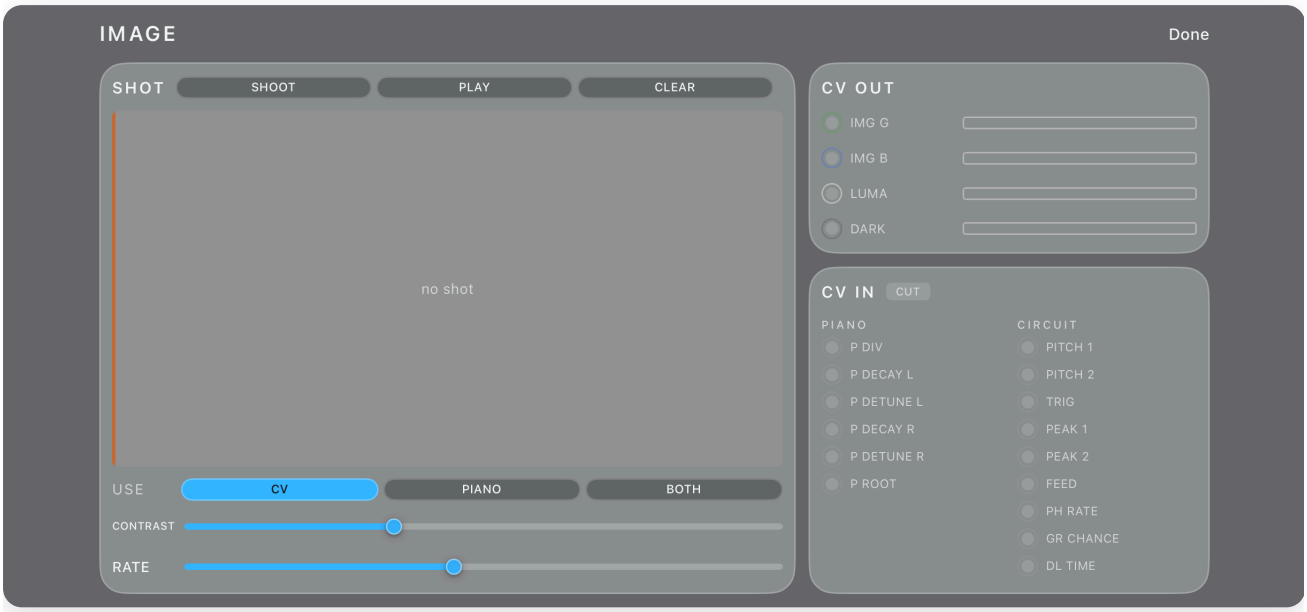

## INPUT · OUTPUT

Choose the microphone and the output — **DEVICE**, **SPEAKER** or **BT**. **AUTO** follows whatever iOS picks; **REFRESH** looks again after you plug something in. RATE and BUFFER show what the system is running at.

## PITCH

Two pitch shifters on the decks alone — the piano and the pad go round them, so a chord stays where you put it. Both are added to the sound rather than replacing it, so they sit as a chord. **PITCH** runs twelve semitones each way; **SEMI** snaps to semitones, **FREE** leaves it where your finger is, **ZERO** returns it. **LEVEL** is how much of that voice is added.

# A red line over one picture IMAGE




**SHOOT** (or the camera key) takes one frame with the back camera and makes it the ground of the app. It is never written to disk — close the app and it is gone. **PLAY** sends a red line across the picture, left to right, in anything from half a second to a minute (**RATE**). **CLEAR** drops it.

Each column the line crosses is read as voltages: its green and blue cast, its brightness, and the inverse — **IMG G**, **IMG B**, **LUMA**, **DARK**. Brightness is taken out of the colours first, so they are not copies of each other; **CONTRAST** widens the difference.

**USE**: **CV** sends the reading to the holes, **PIANO** turns it straight into scale steps, **BOTH** does each.

**CV IN** lists every destination the picture can drive — the piano's DIV, DECAY, DETUNE and ROOT on each side, and on the circuit PITCH 1/2, TRIG, PEAK 1/2, FEED, the phaser rate, grain chance and delay time. Arm a colour under CV OUT, tap a destination, tap again to deepen, once more to pull it out.

11 · PRESETS AND RECORDING

# Keeping what you found

### Presets

Eight slots on the right of the front. Tap one to recall it; hold to store what is on the front now. A slot keeps more than knob positions: it remembers which file each deck was reading, and copies any recording of its own. The bin empties a slot — press it once to arm it, then the number — and that also deletes the slot's recording. The arrow undoes.

### Recording

The microphone key records straight into a deck — take one into deck one, take two into deck two, then round again. While it records the instrument goes quiet so the microphone does not hear itself. Twelve seconds at most, and the recording becomes the waveform the reader runs over: record it, play it, and record that.

12 · VERSIONS

# Where it is, and what comes next

CURRENT

1.1.2

iPhone, landscape · iOS 18.0 or later. This guide describes this version.

NEXT UPDATE

1.2.0

iPhone and iPad, landscape · iOS 18.0 or later. In preparation.

## Changes in 1.2.0

| | |
|---|---|
| **iPad** | Tīwesdæg runs on iPad, in landscape and full screen. The whole instrument fills the display, so every ring and label is larger. |
| **Panels on iPad** | Every panel opens full screen with the same layout as on iPhone, scaled up to fill it. |

Details may change before release. The update will be announced in the newsletter.

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TĪWESDÆG

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